

Azure Load Balancer Outbound Rules and SNAT Planning

Project Requirements

#	Requirement	Description (What it is)	Purpose in Project (Why you're using it)	Free / Paid	Free Alternative
1	Azure Subscription with Permissions	An account-level container in Azure that lets you create and pay for cloud resources, governed by RBAC roles.	Serves as the foundation/environment where every resource for this exercise (LB, VMs, VNet, NSG) is created.	Free	N/A — already free
2	Resource Group	A logical container used to group related Azure resources for unified management.	Keeps all SNAT-project resources bundled together so they can be monitored and deleted as one unit after testing.	Free	N/A — already free
3	Virtual Network (VNet) & Subnet	Azure's private networking construct that lets resources communicate securely within a defined IP address space.	Provides the internal network the backend VMs sit in, and the path traffic takes on its way out through the Load Balancer.	Free	N/A — already free
4	Standard SKU Public IP Address	A publicly routable IP address, available in Basic or Standard SKU, used as an entry/exit point for Azure resources.	Acts as the frontend IP that the Load Balancer uses to route inbound requests and (via outbound rules) manage SNAT for outbound traffic.	Paid	Deallocate/delete the IP right after testing to minimize charges — no cheaper substitute works with Standard LB.
5	Standard SKU Load Balancer	A Layer-4 traffic distribution service that spreads incoming connections across backend resources and supports advanced outbound rule configuration.	The main resource under study — its outbound rules and SNAT port allocation behavior is exactly what this project is testing and documenting.	Paid	No functional substitute — Basic SKU doesn't support outbound rules. Deploy only during active testing and delete afterward.
6	Backend Pool (VMs or VMSS)	A set of virtual machines or a scale set that sits behind the Load Balancer and receives/generates traffic.	Runs the load-generation script that creates outbound connections, actively consuming SNAT ports so their allocation/exhaustion can be observed.	Paid	Use B1s burstable VMs or the Azure Free Tier's 750 free hours/month (if eligible).
7	Network Security Group (NSG)	A firewall-like resource that filters inbound/outbound traffic to subnets or NICs based on rules.	Restricts traffic to only the test script and monitoring connections, keeping the test environment controlled and secure.	Free	N/A — already free
8	Azure Monitor + Diagnostic Settings	A monitoring platform that collects metrics/logs from Azure resources and can route them to various destinations.	Captures SNAT-specific metrics (Used/Allocated SNAT Ports, Connection Count) so you can measure and graph SNAT exhaustion as load increases.	Free (metrics view)	View metrics directly in the Azure Monitor Metrics blade instead of sending logs to Log Analytics.

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9	Scalable Backend Pool (manual/autoscale)	A capability (manual resizing or autoscale rules) to change the number of instances in a pool dynamically.	Lets you increase VM count mid-test to see how SNAT port usage per VM changes as the pool scales, simulating real-world load spikes.	Free (feature) / Paid (extra VM cost)	Scale manually with 1–2 extra B1s VMs instead of configuring full autoscale across many instances.